



Portfolio constructions in cryptocurrency market: A CVaR-based deep reinforcement learning approach[☆]

Tianxiang Cui^a, Shusheng Ding^{b,*}, Huan Jin^a, Yongmin Zhang^b

^a School of Computer Science, University of Nottingham Ningbo, China

^b School of Business, Ningbo University, China

ARTICLE INFO

JEL classification:
G11

Keywords:
Cryptocurrency market
Portfolio optimization
Efficient frontier
Reinforcement learning

ABSTRACT

Cryptocurrency markets have much larger tail risk than traditional financial markets, and constructing portfolios with such large tail risk assets would be challenging. Therefore, cryptocurrency funds demand new superior risk management models and Conditional Value at Risk (CVaR) is a prevailing risk measure for constructing portfolios in stock markets with large tail risk. Consequently, our paper contributes to the literature by developing a new cryptocurrency portfolio model framework based on the CVaR risk measure and a deep reinforcement learning optimization framework. We use the data from cryptocurrency market starting 2015 to 2021, unfolding that CVaR measure with deep learning outperforms the traditional portfolio construction technique. Compared with traditional economic parameter-based portfolio models, our model free based approach can capture the nonlinear compounding effect of multiple risk shocks by deep reinforcement learning on the risk distribution with economic structural breakdown. It can guide investments in financial markets with high tail risks.

1. Introduction

Cryptocurrencies, especially Bitcoin, have attracted much attention due to their innovative features and sharply increased prices (Easley et al., 2019; Prat and Walter, 2021; Liebi, 2022), especially after the outbreak of COVID-19 (Guo et al., 2021b; Mariana et al., 2021; Singh, 2021). As a result, tremendous investors surged into the cryptocurrency market to grasp investment opportunities. Regarding investment, constructing portfolios is a decent and rational way of investing in the financial market (Goetzmann and Kumar, 2008; Bojaj et al., 2022). The theory of portfolios originated from Markowitz (1952), suggesting that the mean and variance of portfolio returns would be the main concerns of investors. Indeed, optimizing the mean and variance of portfolios continues to be the workhorse in the financial research area. As the breakout of COVID-19 and increased global economic policy uncertainty, the demand for investors to seek a safe haven increased. Therefore, the funds in the cryptocurrency market require updated and robust risk management models under the new unprecedented

economic environment, which makes the understanding of the cryptocurrency market even pivotal (Charfeddine et al., 2020; Shahzad et al., 2020; Fousekis and Tzaferi, 2021; Jiang et al., 2021; Qin et al., 2021).

According to Markowitz (1952), the optimization of portfolios rests on two dimensions. The first approach is minimizing the risk of portfolios as the expected return has been given. The second approach is maximizing the expected return as the risk of portfolios has been given. Those optimized portfolios deliver a significant result, which constructs an efficient frontier of portfolios. However, which risk measure is more appropriate and efficient in constructing portfolios in the cryptocurrency market has no unanimous answer. It is arguable that the cryptocurrency market is a fledgling financial market, and thus sound risk measures shall be identified for the market. Volatility and risk measures are key concerns of portfolio managers (Qureshi et al., 2017; Ma and Tang, 2019). The literature maintains that investment in the cryptocurrency market suffers from large tail-risk exposure (Borri, 2019).

[☆] Acknowledgment: We are indebted to Editor Sushanta Mallick, the full professor in Queen Mary University of London, the Guest editor, and two anonymous referees for their contributions during the revision process of Economic Modelling.

This paper is supported by the Special Research Project for the 15th Zhejiang CCP Congress “Digital Technology for Risk Management and Regulations”, China and Academy of Longyuan Construction Financial Research Grant, China (Grant Number : LYZDB2004), Ningbo University and it is also supported by International Research Center for Sustainable Finance, Ningbo University, China.

* Corresponding author.

E-mail addresses: tianxiang.cui@nottingham.edu.cn (T. Cui), dingshusheng@nbu.edu.cn (S. Ding), huan.jin@nottingham.edu.cn (H. Jin), zhangyongmin@nbu.edu.cn (Y. Zhang).

<https://doi.org/10.1016/j.econmod.2022.106078>

Received 25 January 2022; Received in revised form 4 October 2022; Accepted 7 October 2022

Available online 22 October 2022

0264-9993/© 2022 Published by Elsevier B.V.

Traditional risk measures such as variance or VaR might not be reliable in such an emerging market, since they are unable to capture tail risk events. Therefore, we adopt the CVaR risk measure for cryptocurrency portfolio optimization because the CVaR is capable of measuring tail risk. Moreover, when the economic environment is in a turmoil episode, portfolio construction may need to include assets from different asset classes, including stocks and cryptocurrency markets with large tail risk. However, the cryptocurrency market is trading 7 days a week with 24 h per day, which does not share the same trading period with the stock market. As a result, we only consider cryptocurrency assets in this paper to provide a clear understanding for investors to construct portfolios with tail risk. We enable investors to envisage the averaged maximum loss level they may incur, and they can choose a portfolio to balance between the averaged maximum loss level (measured by CVaR) and the expected returns for different portfolio constructions in our efficient frontier.

The main ingredients for constructing the CVaR-based efficient frontier of cryptocurrency portfolios are computation of CVaR and portfolio optimization algorithm. However, estimation error might be one key obstacle to deteriorating portfolio performance (DeMiguel et al., 2009; Olivares-Nadal and DeMiguel, 2018). Copious extant research (Best and Grauer, 1991; Chopra and Ziemba, 1993) unravels the fact that parameter perturbation might be one of the key causes of estimation errors, and thereby, parameter estimations play important roles in model forecasting accuracy. The inaccurate CVaR may exacerbate the loss level, which may lead to injudicious investment decisions. Therefore, it would be essential to propose a new robust method to estimate CVaR. Our approach does not require the specified form of distribution, and we use historical data to simulate return distribution. The copula-based approach has been utilized to generate scenarios using estimated distribution (Kaut and Wallace, 2011). The copula function aids our study in incorporating the correlation among cryptocurrencies. Market estimation errors can thereby be substantially lessened, especially during fluctuating episodes.

Due to the rise of advanced technology, new techniques have been applied in the financial area. For example, genetic programming applications in volatility forecasting (Ding et al., 2019; Ding et al., 2020; Ding et al., 2021), and deep learning (DL) based approach has been applied in the portfolio optimization domain in the past few years (Bai et al., 2022). However, such approaches are undesirable because a large proportion of them are heavily based on price trend prediction models. Since such models often have preset assumptions, which are generally different from practical scenarios, it is necessary to consider using a model-free approach to tackle portfolio optimization problems. Recently, Deep Reinforcement Learning (DRL) approach has been proven effective in policy-based problems due to its remarkable performance in various domains (Atari game (Mnih et al., 2015)). Compared to the DL approach, DRL has two main advantages. Firstly, it is based on a model free approach, which rescues our framework from unrealistic model assumptions. Secondly, DRL-based portfolio optimization can be considered the process of continuously reallocating capital to invest in a number of financial assets based on the latest observations. Therefore, it is natural to utilize DRL to model the continuous asset reallocation process. Hence, we utilize DRL to construct CVaR-based optimal portfolio allocation.

The main contribution of this paper is to propose a new CVaR-based portfolio optimization model for cryptocurrency markets, which employs a DRL methodology. By means of this methodology, we unveil the fact that the efficient frontier formulated by the mean-CVaR model massively dominates the efficient frontier established by the traditional mean-variance model in the cryptocurrency market. Our results contributing to the literature are twofold.

First, we argue that CVaR would be the solid risk measure in the cryptocurrency market, which is capable of capturing tail risk, and tail risk in the cryptocurrency market has been documented, such as jump risk (Scaillet et al., 2020; Zhang et al., 2022a). Additionally,

tail risk can also be observed in other financial markets such as the stock market (Naimoli et al., 2022) and foreign exchange market (Salisu et al., 2022), especially during the turmoil periods such as the COVID episodes (Agarwalla et al., 2021) and during the conflict between Russia and Ukraine (Zhang et al., 2022b). More importantly, tail risk can be propagated across different financial markets during periods of turmoil (Guo et al., 2021a). Therefore, risk management based on CVaR would be extremely helpful in preventing huge losses and tail risk propagation in terms of controlling tail risk not only in the cryptocurrency market but also in other financial markets under external risk shocks.

Second, our DRL-based framework can unbind the preset assumptions of existing models through the model free approach, as those assumptions would be unrealistic during the current turmoil periods. Consequently, establishing the optimized efficient frontier based on the mean-CVaR model could deliver practical investment implications of cryptocurrency in our study. By employing DRL to construct the portfolios, our framework could lessen the model estimation errors because unrealistic assumptions have been unbound. Moreover, existing models often use linear methods to capture risk, which might make it difficult to reflect the compounding effect of tail risks such as the coincidence of Russia-Ukraine conflict and the COVID effect. On the other hand, our framework can capture such an effect by using the model free approach with deep reinforcement learning on the risk distribution and economic structural breakdown impacted by current crisis events. On the basis, we further perform out-of-sample back testing to verify that the CVaR risk measure outperforms the variance risk measure.

The remainder of this paper is structured as follows. Section 2 describes two portfolio optimization models. Section 3 presents the methodology and the data. Section 4 exhibits the empirical results with efficient frontiers in the cryptocurrency market. We adopt deep reinforcement learning to construct the portfolios and perform out-of-sample back testing to verify that the CVaR risk measure outperforms the variance risk measure. Section 5 concludes the paper.

2. Portfolio optimization modelling

2.1. Mean-variance model

The landmark work of Markowitz on the foundation of Modern Portfolio Theory (MPT) has been published in the 1950s (Markowitz, 1952). Markowitz used the mean value that deviates from the expected return to measure portfolio risk. On this basis, the risk level is represented by the covariance between all pairs of assets. Therefore, the basic Markowitz mean-variance model can be considered a quantitative framework by minimizing the portfolio risk given a predetermined expected return level.

The basic mean-variance ignores massive real-world trading constraints. The problem quickly becomes more complex with the additional constraints involved. Some commonly used trading constraints include the transaction costs (Konno and Wijayanayake, 2001), the cardinality constraint (Metaxiotis and Liagkouras, 2012), the bounding constraints (Metaxiotis and Liagkouras, 2012), the minimum trading size (Crama and Schyns, 2003), and the round lot constraint (Mansini et al., 2014). Typically, when such real-world trading constraints are added to extend the basic mean-variance model, integer variables are needed. In this work, we mainly investigate two constraints, namely, the cardinality and the bounding constraints. Mathematically, the extended cardinality constrained mean-variance model can take the following format (see Chang et al., 2000):

$$\min_{w_i} \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} \quad (1)$$

$$\text{subject to } \sum_{i=1}^n w_i \mu_i = R^* \quad (2)$$

$$\sum_{i=1}^n w_i = 1 \quad (3)$$

$$\sum_{i=1}^n s_i = K \quad (4)$$

$$\epsilon_i s_i \leq w_i \leq \delta_i s_i, \quad i = 1, \dots, n \quad (5)$$

$$s_i \in \{0, 1\}, \quad i = 1, \dots, n \quad (6)$$

where K is the cardinality constraint and ϵ_i and δ_i are the bounding constraints, respectively.

The objective function (1) uses the covariance matrix (positive semidefinite) to minimize the volatility (variance) associated with the chosen portfolio. Eq. (2) defines the portfolio expected return, R^* , which meets the investor's preference. Eq. (3) is the budget constraint, which ensures that the assets' weights sum to one. Eq. (4) is the cardinality constraint, which ensures that exactly K assets are chosen to be held in the portfolio. Eq. (5) is the buy-in threshold bounding on asset investments. If $s_i = 0$, asset a_i is not chosen to be selected in the portfolio, and the associated weight w_i is also 0. If $s_i = 1$, then asset a_i is chosen to be selected in the portfolio, and the associated weight should lie in $[\epsilon_i, \delta_i]$, where $0 \leq \epsilon_i \leq \delta_i \leq 1$. Eq. (6) is the integrality constraint, reflecting the inclusion or exclusion of a particular asset.

2.2. Mean-CVaR model

The classical Markowitz mean-variance portfolio optimization model relies on perfect market information (i.e., assets' expected returns and their variance-covariance matrix are assessed by financial analysts with perfect precision). In practice, this assumption is often unrealistic, and it is complicated even to obtain accurate estimates for them, since there are massive possible sources of error that can affect the estimation. In order to construct portfolios that are less impacted by inaccuracies in the estimations of the mean and the covariance matrix of returns, market uncertainty becomes one key consideration. In this section, we investigate market probabilistic uncertainties and construct cryptocurrency portfolios with minimization of tail risk, which is characterized by conditional VaR (CVaR).

2.2.1. Value at Risk (VaR)

Practically, possibilities of huge loss would be the key concern of investors, and thus, they are enthusiastic to lower it. Value at Risk (VaR) (Jorion, 2006; Pritsker, 1997) is proposed to give the maximum possible loss α with a specified confidence level β . In other words, the probability of the portfolio loss exceeding the threshold α is $1 - \beta$.

Mathematically, define $f(x, \xi)$ as the loss function. The decision vector is $x \in \mathbb{R}^n$, and the uncertain vector is $\xi \in \mathbb{R}^m$. Denote $p(\xi)$ as the probability density distribution of ξ . Therefore, the probability of the loss function being less than or equal to the threshold α can be represented as:

$$\Psi(x, \alpha) = \int_{f(x, \xi) \leq \alpha} p(\xi) d\xi \quad (7)$$

Given the specified probability β in $(0, 1)$, denote $\alpha_\beta(x)$, as the β -VaR for the loss random variable associated with x :

$$\alpha_\beta(x) = \min\{\alpha \in \mathbb{R} : \Psi(x, \alpha) \geq \beta\} \quad (8)$$

2.2.2. Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR) was introduced by Rockafellar and Uryasev, 2000. Compared with VaR, CVaR is more consistent due to its subadditivity and convexity, and it is a coherent risk measure (Pflug, 2000). CVaR gives the mean value of the losses that are larger than the VaR value. Mathematically, it is defined as follows:

$$\phi_\beta(x) = (1 - \beta)^{-1} \int_{f(x, \xi) \geq \alpha_\beta(x)} f(x, \xi) p(\xi) d\xi \quad (9)$$

If we replace $\alpha_\beta(x)$ with an analytical representation, we can have the following equivalent formulation:

$$F_\beta(x, \alpha) = \alpha + (1 - \beta)^{-1} \int_{f(x, \xi) \geq \alpha} (f(x, \xi) - \alpha) p(\xi) d\xi \quad (10)$$

Rockafellar and Uryasev, 2000 proved that $F_\beta(x, \alpha)$ is convex with respect to α . The VaR is the value of α at which $F_\beta(x, \alpha)$ obtains the minimum value. The corresponding minimum value is called CVaR.

2.2.3. Minimizing CVaR

β -CVaR would always be not less than β -VaR for a specified probability level β . In fact, CVaR can be optimized by minimizing the function $F_\beta(x, \alpha)$, and the corresponding VaR can be obtained at the same time (Uryasev, 2000). If (x^*, α^*) is the solution for minimizing $F_\beta(x, \alpha)$, then $F_\beta(x^*, \alpha^*)$ would be the value of optimal CVaR, and α^* is the associated VaR value accordingly. If we introduce an auxiliary function $Z(\xi)$ ($Z(\xi) \geq f(x, \xi) - \alpha$ and $Z(\xi) \geq 0$), the equivalent formulation can be represented as:

$$F_\beta(x, \alpha) = \min_{Z(\xi)} \alpha + (1 - \beta)^{-1} E(Z(\xi)) \quad (11)$$

$$\text{and } \beta - CVaR(x) = \min_{\alpha, Z(\xi)} \alpha + (1 - \beta)^{-1} E(Z(\xi)) \quad (12)$$

s.t.

$$Z(\xi) \geq f(x, \xi) - \alpha$$

$$Z(\xi) \geq 0$$

$$\alpha \in \mathbb{R}$$

In this work, the future uncertain cryptocurrency prices are represented using ξ . In practice, it is not possible to have the analytical representation of $p(\xi)$. Alternatively, we use copula-based methods (Høyland and Wallace, 2001; Høyland et al., 2003) to generate scenarios based on historical observations of each cryptocurrency price. Suppose $y_1, y_2, \dots, y_N$ are N scenarios obtained from the density $p(\xi)$, $F_\beta(x, \alpha)$ can therefore be computed as:

$$F_\beta(x, \alpha) = \alpha + (1 - \beta)^{-1} \sum_{i=1}^N p_i (f(x, y_i) - \alpha)^+ \quad (13)$$

where the portfolio loss function of scenario i is $f(x, y_i)$. p_i is the probability of scenario i and $(f(x, y_i) - \alpha)^+ = \max(0, (f(x, y_i) - \alpha))$. By using the auxiliary function $Z(\xi)$, we can obtain the auxiliary variable z_i , where $z_i \geq f(x, y_i) - \alpha$, $z_i \geq 0$, $i = 1, \dots, N$. Suppose R^i is the portfolio return on scenario i and R^* is the expected return level predetermined by the investor. We simplify the minimization of the function $F_\beta(x, \alpha)$ as:

$$\min \quad \alpha + (1 - \beta)^{-1} \sum_{i=1}^N p_i z_i \quad (14)$$

s.t.

$$z_i \geq f(x, y_i) - \alpha \quad i = 1, \dots, N$$

$$z_i \geq 0 \quad i = 1, \dots, N$$

$$\sum_{i=1}^N p_i R^i \geq R^*$$

$$\alpha \in \mathbb{R}$$

$$x \in \mathbb{R}^n$$

where β is the confidence level and $f(x, y_i)$ is the loss function.

It has been proven that such a formulation is more stable in various scenarios (Andersson et al., 2001; Krokhmal et al., 2002). Because of the different portfolio construction mechanisms, we establish the mean-variance portfolios according to Eqs. (1)–(6), and we establish the mean-CVaR portfolios according to Eq. (14).

3. Methodology

3.1. Reinforcement learning

Generally, reinforcement learning attempts to maximize the total reward, which has been set at the beginning of the problem. The agent (decision maker) optimizes its policy for choosing the proper action at

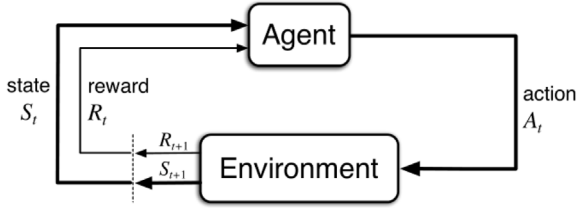


Fig. 1. Agent-Environment Interaction.

different states by interacting with the environment to maximize the total rewards, as shown in Fig. 1.

Formally, an MDP can be formulated as a tuple $M = (S, A, R_a, P_a, \gamma)$, where:

- S is the set of state representations.
- A is the set of actions that can be selected by the agent.
- R_a is the immediate reward obtained after transferring the current state to the next state due to action a .
- P_a is the state transition probability, which can be represented as $P_a(s', s) = \Pr(s_{\tau+1} = s' | s_\tau = s, a_\tau = a)$, indicating the probability of action a in state s at time instance τ leading to state s' at time instance $\tau + 1$.
- $\gamma \in [0, 1]$ represents the decay factor.

The purpose of the agent to act in MDP aims to find an optimal policy π_θ that can map states into actions that can maximize the discounted total rewards.

3.2. The proposed RL approach

In this work, we adopt the OpenAI default RL algorithm, Proximal Policy Optimization (PPO) (Schulman et al., 2017), to address the two portfolio optimization models described in Section 2. Normally, in the RL training process, a guaranteed monotonic improvement is favored. Large step sizes can hurt the gradient descent performance, while small steps can result in slow convergence. PPO computes an update at each step that minimizes the cost function while keeping the relatively small differences between the new policy and the old policy and requires small efforts in parameter tuning. PPO uses a clipped objective function to penalize parameter updates to ensure that the new policy obtained is not far from the previous policy.

3.3. Data

The sample data we collect are for Bitcoin, Ripple, Litecoin, Ethereum, Tether (USDT) and Dash from the Yahoo Financial database on a daily basis. The sample covers the period from 6 August, 2015 to 6 August, 2021 (2199 daily total observations). Since we are using the RL method, we thus divided the sample into two parts. We use the sample ranging from 6 August, 2015 to 31 December, 2019 as the input, severing the training set and the sample starting from 1 January, 2020 to 6 August, 2021 as the testing period. We report all correlations in Table 1. The highest correlation appears between Bitcoin and Ethereum, which is 0.9092, while the lowest correlation is between Bitcoin and Tether, which is -0.2818 . This indicates that the key role Bitcoin plays in the cryptocurrency market. We plot all six types of cryptocurrencies in Fig. 2. The prices of Tether are approximately 1 USD, and other cryptocurrencies exhibit similar moving trends. In particular, the cryptocurrency named Tether is a kind of stable coin, which is pegged to the US Dollar, and Tether is anchored at 1 USD. Stable coins serve as a medium to price cryptocurrencies in dollars without a US Dollar bank account (Wei, 2018).

Table 1

Pearson correlation coefficients for Bitcoin, Dash, Ethereum, Litecoin, Tether and Ripple.

Cryptocurrency	Bitcoin	Dash	Ethereum	Litecoin	Tether	Ripple
Bitcoin	1.0000					
Dash	0.8165	1.0000				
Ethereum	0.9092	0.8003	1.0000			
Litecoin	0.9077	0.9495	0.8821	1.0000		
Tether	-0.2818	-0.2071	-0.2443	-0.2548	1.0000	
Ripple	0.7573	0.8591	0.8586	0.8621	-0.1400	1.0000

In addition, we also present returns of all six cryptocurrencies in Fig. 3. It is observable that returns of Tether are quite stable compared with the other five cryptocurrencies and that Bitcoin has a more dispersed return distribution. To optimize the mean-variance model, we calculate the covariance between each individual cryptocurrency based on the dataset, and to optimize the mean-CVaR model, massive scenarios have been created from the dataset. Those scenarios are created by using the t-copula, which could create different return scenarios for single return scenario cryptocurrency such as Tether. The histogram of each cryptocurrency scenario can be found in Fig. 4.

3.4. RL encoding

State: The state s_τ at time instance τ is defined as a m by $4n$ matrix. Each row of the matrix is a $4n$ -dimensional vector $[o', h', l', c']$, where $o' = [o'_0, o'_1, o'_2, \dots, o'_n]^T$, representing the opening price of each cryptocurrency at the time step t , $h' = [h'_0, h'_1, h'_2, \dots, h'_n]^T$, representing the high price of each cryptocurrency at the time step t , $l' = [l'_0, l'_1, l'_2, \dots, l'_n]^T$, representing the low price of each cryptocurrency at the time step t , $c' = [c'_0, c'_1, c'_2, \dots, c'_n]^T$, representing the closing price of each cryptocurrency at the time step t and n is the total number of cryptocurrencies. The column m of the matrix represents the total trading days (time window).

Action: Given the state s_τ , the actions a_τ at time instance τ can be defined as an n -dimensional vector $a_\tau = [w'_1, w'_2, w'_3, \dots, w'_n]^T$, representing weight distributions of the portfolio at each time instance for a given predefined return level R^* . Thus, the action space will be continuous.

Reward: The reward R_t is set to be the negative value of two objective functions, Eq. (1) for the mean-variance optimization model, where the covariance between cryptocurrency i and cryptocurrency j is calculated as $\sigma_{ij} = \frac{1}{m} \sum_{t=1}^m (x'_t(i) - \mu(i))(y'_t(j) - \mu(j))$, and Eq. (14) for the mean-CVaR optimization model. The special case that $i = j$ is for the variance estimation. At each time instance, the RL agent employs a policy π to interact with the environment. The agent's goal is to explore a policy that can maximize the cumulative discounted reward as calculated by $\sum_{t'=0}^T \gamma^{t'-\tau} r_{t'}, \forall \tau \in [1, T]$ and $\gamma \in [0, 1]$.

Network Structure: Since our state is represented by a matrix, the deep residual network (He et al., 2016) is adopted for approximating the policy. The procedure of generating the weights and other parameters is based on the state, which is the observation to the machine, and the feedback of the parameter fitting, which is based on the reward feedback system. We use the full sample as the input, which indicates that all results are in-sample results.

4. Empirical results

4.1. Efficient frontier based on the mean-CVaR model

According to Pareto optimality (Fonseca and Fleming, 1995), all portfolios on the efficient frontier are optimized results from our model since they demonstrate the best possible trade-off of risk levels against return levels. We use the methodology of minimizing CVaR mentioned in Section 3.2 to yield optimized portfolios establishing the efficient frontier. The efficient frontier of cryptocurrency portfolios using the mean-CVaR model is presented in Fig. 5.

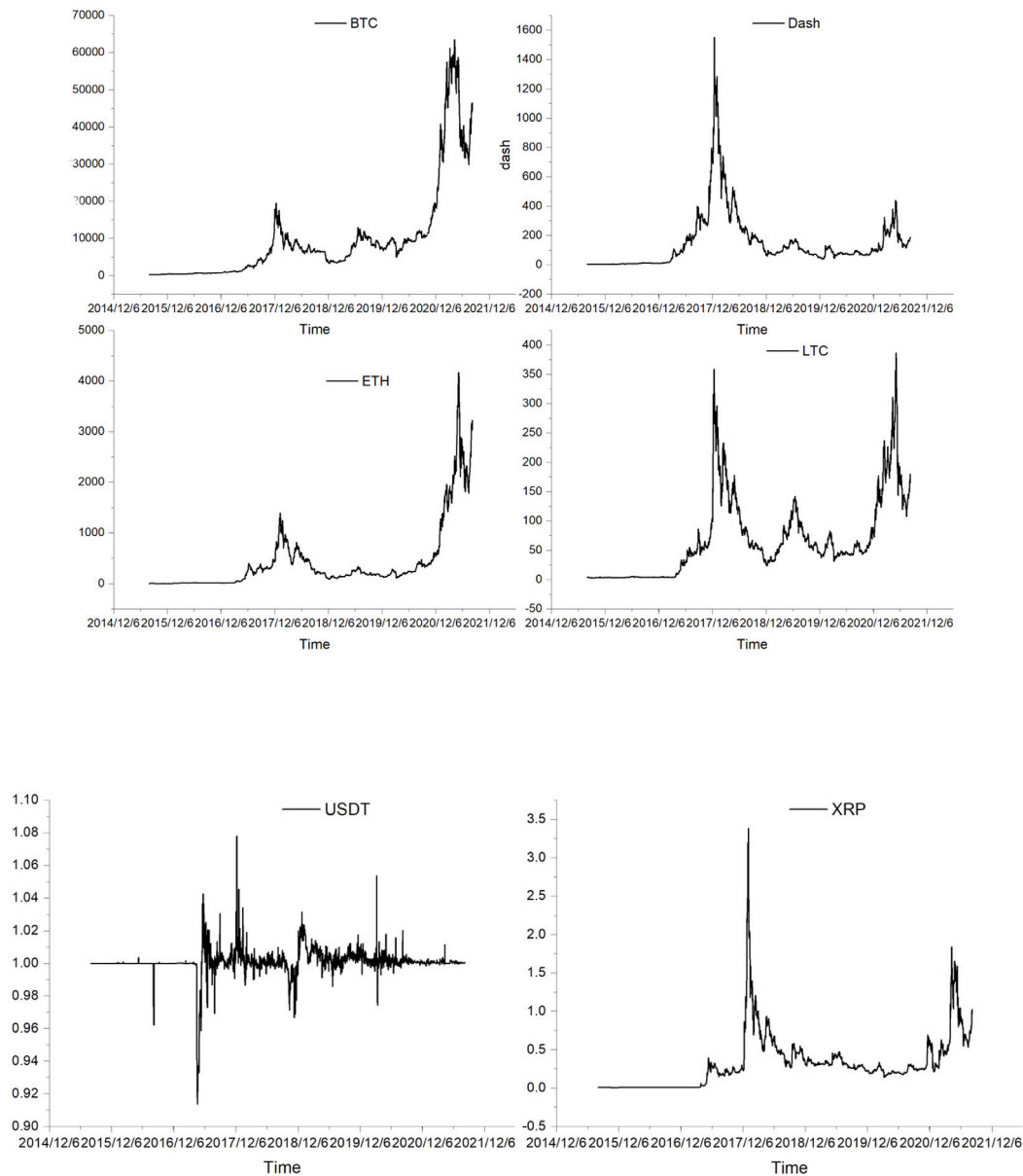


Fig. 2. Plot of 6 cryptocurrencies' prices.

We further select ten portfolios on the efficient frontier as representative portfolios. Those ten (2*5) portfolios are extracted on the efficient frontiers in Fig. 8, ranging from smallest CVaR to largest CVaR for both mean–variance and mean–CVaR curves. To obtain representative portfolios, the distance between any two near portfolios is the same on the efficient frontiers. Fig. 6 presents different corresponding weights of cryptocurrencies in those portfolios. The figure also shows different CVaRs of ten portfolios, where the portfolio with the smallest CVaR is numbered Portfolio 1 and the portfolio with the largest CVaR is numbered Portfolio 10. From Fig. 6, the portfolio with the smallest CVaR is constituted with approximately 50% weight of Bitcoin, suggesting that Bitcoin can help to stabilize the portfolio return similar to the function of gold (Klein et al., 2018). On the other hand, the portfolio with the largest CVaR is constituted with approximately 100% weight of Dash, indicating that Dash might be a considerably volatile cryptocurrency since it formulates the riskiest portfolio on the efficient frontier.

4.2. Comparison between mean–CVaR and mean–variance portfolio optimization

This subsection compares the optimized results of mean–CVaR with the mean–variance model by undertaking a two-stage estimation. First, we adopt our scenario-based optimization model to find the efficient frontier regarding Markowitz's mean–variance model. Then, we calculate the CVaR of portfolios on the mean–variance efficient frontier, and we present this efficient frontier (dashed line) with the previously estimated mean–CVaR efficient frontier (solid line) in Fig. 7. It is clear that the mean–CVaR efficient frontier heavily dominates the mean–variance efficient frontier. For example, from a horizontal view, the mean–CVaR model has a much lower CVaR for the 0.4% return compared with the mean–variance model. From a vertical view, the mean–CVaR model has a much higher return for the CVaR of 0.12 compared with the mean–variance model.

Furthermore, we show the different weights of ten portfolios on both frontiers (see Fig. 8). This figure presents portfolios according

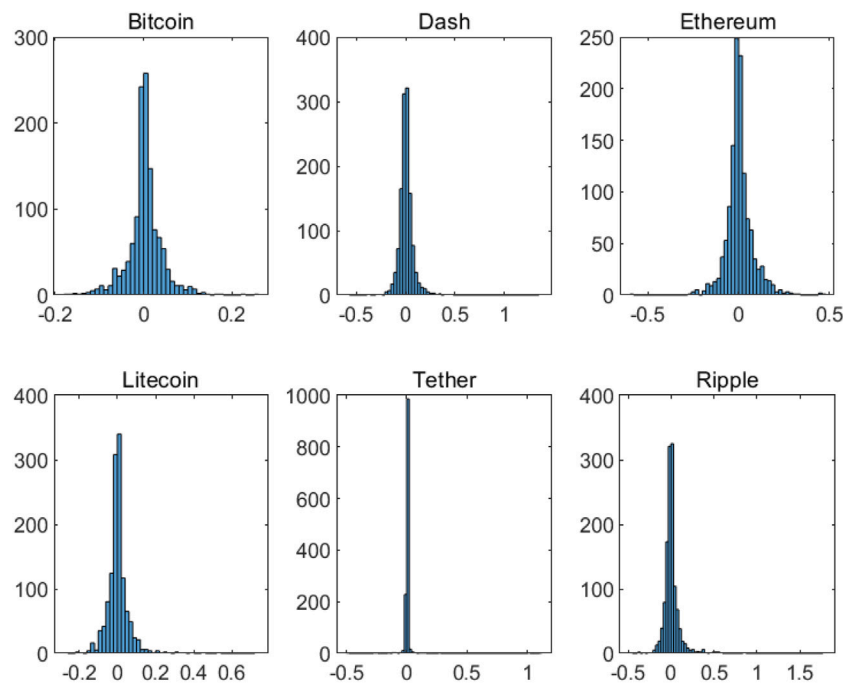


Fig. 3. Histogram of 6 cryptocurrencies' returns.

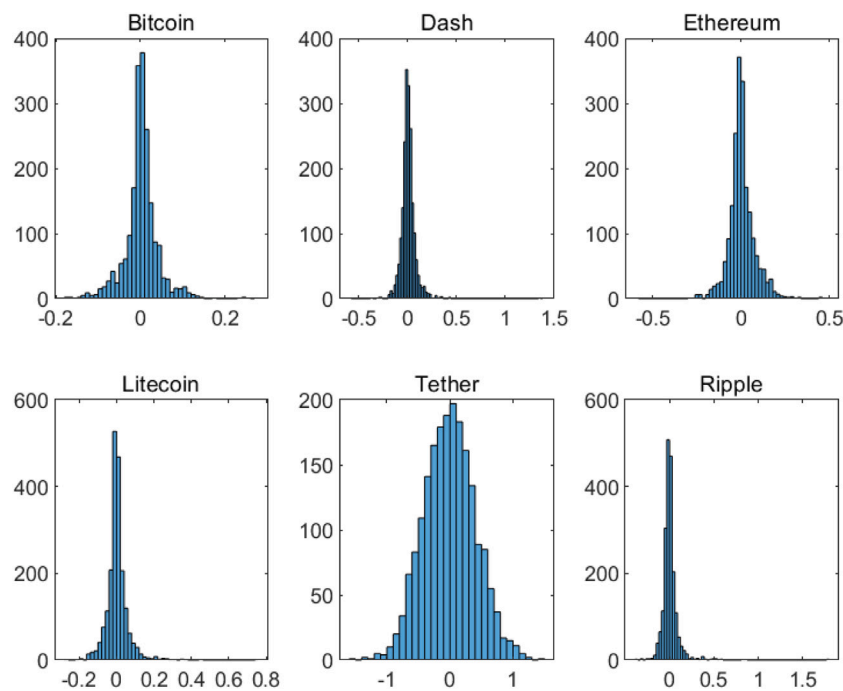


Fig. 4. Histogram of cryptocurrency scenarios.

to their CVaRs (portfolio with small CVaR is on the left). It also exhibits a number of interesting characteristics of the two models regarding portfolio construction in the cryptocurrency market. First, the proportion and distribution of Bitcoin are quite similar in both models, suggesting that Bitcoin can stabilize the portfolio since it disappears in portfolios on the right side. Regardless of the risk measure, Bitcoin is an essential part of less risky cryptocurrency portfolios. Second, the proportion of Tether is divergent in the two models. Less risky cryptocurrency portfolios generated by the mean-variance model largely rely on the composition of Tether. On the other hand, however, Tether

contributes comparatively little proportion in portfolios constructed by the mean-CVaR model. In contrast, Litecoin has little contribution in mean-variance portfolios, whereas it has a higher proportion in mean-CVaR portfolios. It is arguable that less risky cryptocurrency portfolios shall comprise Litecoin rather than Tether to keep portfolios efficient. Finally, the riskiest portfolio is constructed by Dash in the mean-CVaR model, while it is constructed by Ripple under the mean-variance framework. Therefore, if an investor intends to invest all money into one cryptocurrency, Dash might be a better choice than Ripple since the pure Dash constructed portfolio is efficient.

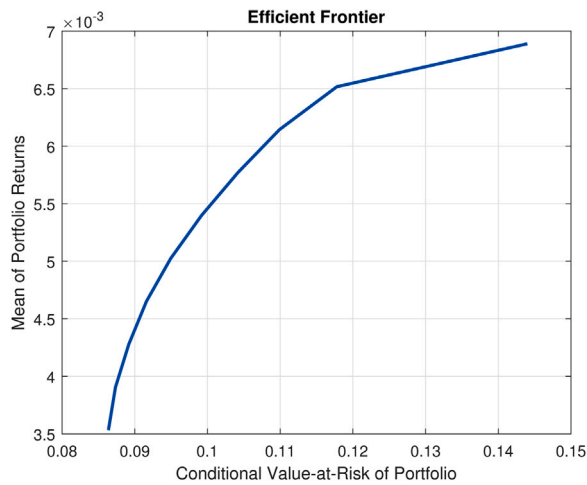


Fig. 5. Efficient frontier of a cryptocurrency portfolio using the mean-CVaR model.

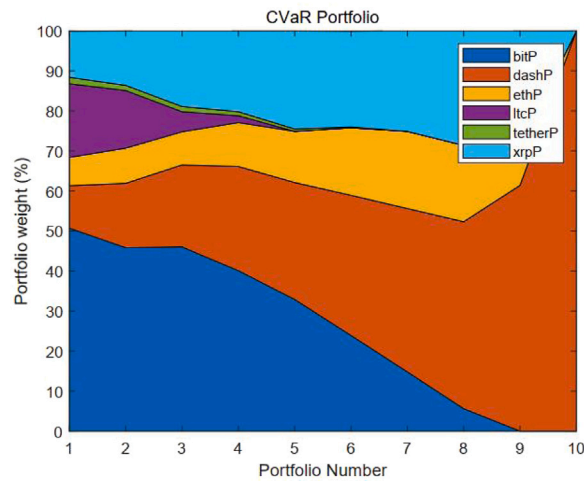


Fig. 6. Cryptocurrencies' weights of different portfolios.

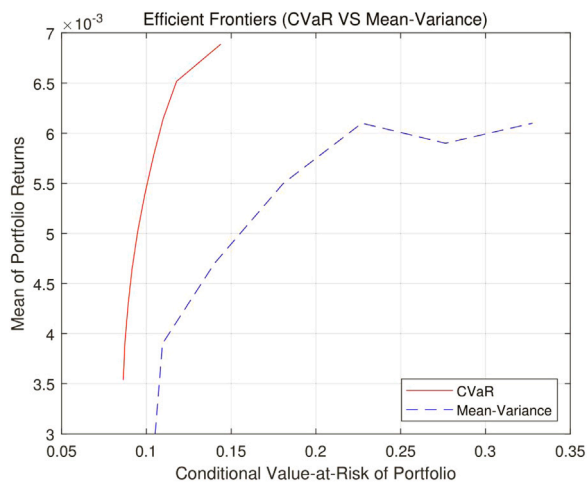


Fig. 7. Comparison of Efficient frontier of mean-CVaR model with mean-variance model.

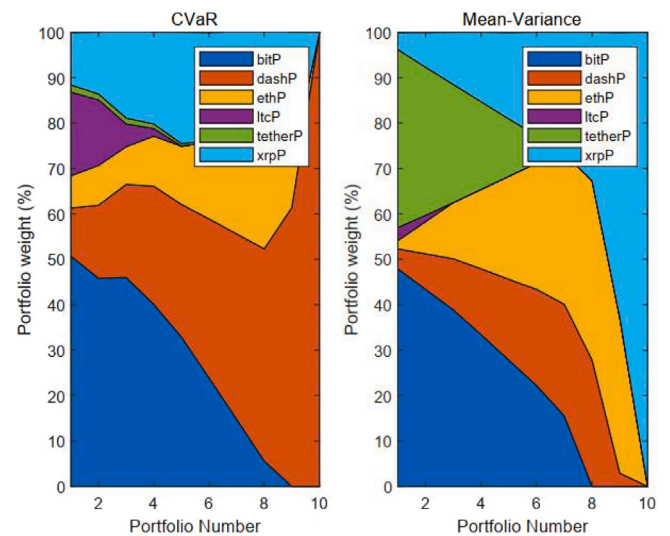


Fig. 8. Comparison of cryptocurrencies' weights of mean-CVaR model with mean-variance model.

4.3. Out-of-sample back testing

In order to further verify that the portfolios on both mean-variance and mean-CVaR efficient frontiers would be favored for CVaR measure. We employ deep reinforcement learning to construct the portfolios according to the mean-variance and mean-CVaR mechanisms for out-of-sample back testing. Based on the reinforcement learning process, we have obtained both portfolios for mean-variance and mean-CVaR during the out-of-sample period. We back-tested the two portfolios to check the performance of the portfolios, and the test periods start from January 1, 2019, to 6, August 2021. Table 2 reports the four moments of ten portfolios, where five portfolios are under the mean-variance framework (labeled MVP) and the other five portfolios are under the mean-CVaR framework (labeled MCP). The ten portfolios are extracted according to Fig. 6 for the weights and different variances and CVaRs, which are evenly distributed in the efficient frontier according to the CVaR horizon. Those ten (2×5) portfolios are extracted on the efficient frontiers in Fig. 7, ranging from smallest CVaR to largest CVaR for both mean-variance and mean-CVaR curves. Therefore, these portfolios are selected such that the first portfolio is at the starting point of the efficient frontier and the fifth portfolio is at the ending point of the efficient frontier with respect to the CVaR values. To have representative portfolios, the distance between any two near portfolios is the same on the CVaR horizon (the X-axis). The evenly distributed portfolio can represent the performance of the model based on different CVaR values.

From Table 2, we can observe that the average returns, which are the first moments, of the mean-variance portfolios are generally lower than those of the mean-CVaR portfolios. More importantly, the standard deviations, which are the second moments, of the mean-variance portfolios are larger than those of the mean-CVaR portfolios. This indicates that the portfolios constructed under the mean-CVaR framework have higher returns with lower risk compared with the portfolios constructed under the mean-variance model. Accordingly, it would be clear that the CVaR risk measure overwhelmingly outperforms the variance risk measure. We also present the annualized return and variance in Table 3, which is consistent with Table 2. Moreover, the third moment, which is the skewness for two groups of portfolios, is almost identical, and the fourth moment, which is the kurtosis, for the mean-CVaR portfolios is closer to 3 compared with the mean-variance portfolios. We thereby demonstrate that the CVaR risk measure performs better than the variance risk measure in the cryptocurrency market.

Table 2

The out-of-sample comparison between mean–variance portfolios and mean-CVaR portfolios regarding the four moments of the portfolios. Portfolios under the mean–variance framework are labeled MVP, and portfolios under the mean-CVaR framework are labeled MCP. The sample period is from January 1, 2019, to August 6, 2021. In this table, 1st M, 2nd M, 3rd M and 4th M represent the first, second, third and fourth moments of each portfolio, respectively.

	MVP1	MVP2	MVP3	MVP4	MVP5	MCP1	MCP2	MCP3	MCP4	MCP5
1 st M	0.00256	0.00256	0.00256	0.00255	0.00199	0.00266	0.00271	0.00275	0.00287	0.00314
2 nd M	0.566	0.573	0.602	0.638	0.514	0.417	0.427	0.446	0.441	0.513
3 rd M	4.72	4.72	4.72	4.73	4.68	4.68	4.68	4.69	4.68	4.68
4 th M	2.38	2.38	2.39	2.4	2.47	2.50	2.51	2.51	2.54	2.97

Table 3

The out-of-sample comparison between mean–variance portfolios and mean-CVaR portfolios regarding annualized return and variance of the portfolios. Portfolios under the mean–variance framework are labeled MVP, and portfolios under the mean-CVaR framework are labeled MCP. The sample period is from January 1, 2019, to August 6, 2021.

	MVP1	MVP2	MVP3	MVP4	MVP5	MCP1	MCP2	MCP3	MCP4	MCP5
Return	0.93	0.93	0.93	0.93	0.73	0.97	0.99	1.00	1.05	1.15
Variance	10.81	10.95	11.50	12.19	9.82	7.97	8.16	8.52	8.43	9.80

5. Conclusion

In summary, we use a scenario-based optimization model incorporating future cryptocurrency price uncertainty to construct portfolios in the cryptocurrency market. This methodology enables us to verify that CVaR is a superior risk measure in constructing portfolios compared with variance concerning the cryptocurrency market. We shall emphasize that the methodology helps us unveil the fact that the efficient frontier formulated by the mean-CVaR model enormously dominates the efficient frontier established by the mean–variance model in the cryptocurrency market.

The contributions of our study are twofold. Firstly, our framework is capable of capturing tail risk, especially during the turmoil periods such as the COVID episodes and during the conflict between Russia and Ukraine. As a result, our framework can be helpful to risk management in preventing huge losses and tail risk propagation in terms of controlling tail risk. Secondly, our DRL-based framework can unbind the preset assumptions of existing models, which could produce practical investment implications of cryptocurrency for investors. Compared with traditional economic parameter-based portfolio models, our model free based approach can capture the nonlinear compounding effect of multiple external risk shocks by deep reinforcement learning on the risk distribution with economic structural breakdown.

In order to provide useful investment implications, we also specify detailed weights of six cryptocurrencies in ten representative portfolios on generated efficient frontiers. To ensure that the portfolios are optimized on both mean–variance and mean-CVaR efficient frontiers. We employ deep reinforcement learning to construct the portfolios according to the mean–variance and mean-CVaR mechanisms. We have demonstrated that the CVaR surpasses variance as the risk measure in the cryptocurrency market during the out-of-sample testing period. Consequently, our framework could be extremely useful to all risk preference investors who intend to construct a cryptocurrency-based portfolio. Our study can provide insights for investors to construct portfolios or design trading strategies in the cryptocurrency market.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

References

- Agarwalla, S.K., Varma, J.R., Virmani, V., 2021. The impact of COVID-19 on tail risk: Evidence from Nifty index options. *Econom. Lett.* 204, 109878.
- Andersson, F., Uryasev, S., Uryasev, S., 2001. Credit risk optimization with conditional value-at-risk criterion. *Math. Program.* 89 (2), 273–291.
- Bai, L., Zheng, K., Wang, Z., Liu, J., 2022. Service provider portfolio selection for project management using a BP neural network. *Ann. Oper. Res.* 308 (1), 41–62.
- Best, M.J., Grauer, R.R., 1991. On the sensitivity of mean-variance-efficient portfolios to changes in asset means: Some analytical and computational results. *Rev. Financ. Stud.* 4 (2), 315. <http://dx.doi.org/10.1093/rfs/4.2.315>.
- Bojaj, M.M., Muhadinovic, M., Bracanovic, A., Mihailovic, A., Radulovic, M., Jolicic, I., Milosevic, I., Milacic, V., 2022. Forecasting macroeconomic effects of stablecoin adoption: A Bayesian approach. *Econ. Model.* 109, 105792.
- Borri, N., 2019. Conditional tail-risk in cryptocurrency markets. *J. Empir. Financ.* 50, 1–19.
- Chang, T.J., Meade, N., Beasley, J.E., Sharaiha, Y.M., 2000. Heuristics for cardinality constrained portfolio optimisation. *Comput. Oper. Res.* 27 (13), 1271–1302.
- Charfeddine, L., Benlagha, N., Maouchi, Y., 2020. Investigating the dynamic relationship between cryptocurrencies and conventional assets: Implications for financial investors. *Econ. Model.* 85, 198–217.
- Chopra, V.K., Ziemba, W.T., 1993. The effect of errors in means, variances, and covariances on optimal portfolio choice. *J. Portf. Manag.* 19 (2), 6–11.
- Crama, Y., Schyns, M., 2003. Simulated annealing for complex portfolio selection problems. *European J. Oper. Res.* 150 (3), 546–571. [http://dx.doi.org/10.1016/S0377-2217\(02\)00784-1](http://dx.doi.org/10.1016/S0377-2217(02)00784-1).
- DeMiguel, V., Garlappi, L., Uppal, R., 2009. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Rev. Financ. Stud.* 22 (5), 1915–1953.
- Ding, S., Cui, T., Zhang, Y., 2020. Incorporating the RMB internationalization effect into its exchange rate volatility forecasting. *North Am. J. Econ. Finance* 54, 101103.
- Ding, S., Cui, T., Zhang, Y., Li, J., 2021. Liquidity effects on oil volatility forecasting: From fintech perspective. *PLoS One* 16 (11), e0260289.
- Ding, S., Zhang, Y., Duygun, M., 2019. Modeling price volatility based on a genetic programming approach. *Br. J. Manag.* 30 (2), 328–340.
- Easley, D., O'Hara, M., Basu, S., 2019. From mining to markets: The evolution of bitcoin transaction fees. *J. Financ. Econ.*
- Fonseca, C.M., Fleming, P.J., 1995. An overview of evolutionary algorithms in multiobjective optimization. *Evol. Comput.* 3 (1), 1–16.
- Fousekis, P., Tzaferi, D., 2021. Returns and volume: Frequency connectedness in cryptocurrency markets. *Econ. Model.* 95, 13–20.
- Goetzmann, W.N., Kumar, A., 2008. Equity portfolio diversification. *Rev. Finance* 12 (3), 433–463.
- Guo, Y., Li, P., Li, A., 2021a. Tail risk contagion between international financial markets during COVID-19 pandemic. *Int. Rev. Financ. Anal.* 73, 101649.
- Guo, X., Lu, F., Wei, Y., 2021b. Capture the contagion network of bitcoin—Evidence from pre and mid COVID-19. *Res. Int. Bus. Finance* 58, 101484.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition. CVPR, pp. 770–778. <http://dx.doi.org/10.1109/CVPR.2016.90>.
- Høyland, K., Kaut, M., Wallace, S.W., 2003. A heuristic for moment-matching scenario generation. *Comput. Optim. Appl.* 24 (2–3), 169–185.
- Høyland, K., Wallace, S.W., 2001. Generating scenario trees for multistage decision problems. *Manage. Sci.* 47 (2), 295–307.
- Jiang, Y., Lie, J., Wang, J., Mu, J., 2021. Revisiting the roles of cryptocurrencies in stock markets: A quantile coherency perspective. *Econ. Model.* 95, 21–34.
- Jorion, P., 2006. *Value At Risk: The New Benchmark for Managing Financial Risk*, third ed. McGraw-Hill Education.

- Kaut, M., Wallace, S.W., 2011. Shape-based scenario generation using copulas. *Comput. Manag. Sci.* 8 (1–2), 181–199.
- Klein, T., Thu, H.P., Walther, T., 2018. Bitcoin is not the new gold—a comparison of volatility, correlation, and portfolio performance. *Int. Rev. Financ. Anal.* 59, 105–116.
- Konno, H., Wijayanayake, A., 2001. Portfolio optimization problem under concave transaction costs and minimal transaction unit constraints. *Math. Program.* 89 (2), 233–250. <http://dx.doi.org/10.1007/PL00011397>.
- Krokhmal, P., Palmquist, J., Uryasev, S., 2002. Portfolio optimization with conditional value-at-risk objective and constraints. *J. Risk* 4, 11–27.
- Liebi, L.J., 2022. Is there a value premium in cryptoasset markets? *Econ. Model.* 105777.
- Ma, L., Tang, Y., 2019. Portfolio manager ownership and mutual fund risk taking. *Manag. Sci.*
- Mansini, R., Ogryczak, W., Speranza, M.G., 2014. Twenty years of linear programming based portfolio optimization. *European J. Oper. Res.* 234 (2), 518–535. <http://dx.doi.org/10.1016/j.ejor.2013.08.035>, 60 years following Harry Markowitz's contribution to portfolio theory and operations research.
- Mariana, C.D., Ekaputra, I.A., Husodo, Z.A., 2021. Are bitcoin and ethereum safe-havens for stocks during the COVID-19 pandemic? *Finance Res. Lett.* 38, 101798.
- Markowitz, H.M., 1952. Portfolio selection. *J. Finance* 7 (1), 77–91.
- Metaxiotis, K., Liagkouras, K., 2012. Multiobjective evolutionary algorithms for portfolio management: A comprehensive literature review. *Expert Syst. Appl.* 39 (14), 11685–11698. <http://dx.doi.org/10.1016/j.eswa.2012.04.053>.
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., Hassabis, D., 2015. Human-level control through deep reinforcement learning. *Nature* 518 (7540), 529–533.
- Naimoli, A., Gerlach, R., Storti, G., 2022. Improving the accuracy of tail risk forecasting models by combining several realized volatility estimators. *Econ. Model.* 107, 105701.
- Olivares-Nadal, A.V., DeMiguel, V., 2018. A robust perspective on transaction costs in portfolio optimization. *Oper. Res.* 66 (3), 733–739.
- Pflug, G.C., 2000. Some remarks on the value-at-risk and the conditional value-at-risk. In: *Probabilistic Constrained Optimization*, vol. 49, Springer US, pp. 272–281.
- Prat, J., Walter, B., 2021. An equilibrium model of the market for bitcoin mining. *J. Polit. Econ.* 129 (8).
- Pritsker, M., 1997. Evaluating value at risk methodologies: Accuracy versus computational time. *J. Financ. Serv. Res.* 12 (2), 201–242.
- Qin, M., Su, C.-W., Tao, R., 2021. BitCoin: A new basket for eggs? *Econ. Model.* 94, 896–907.
- Qureshi, F., Kutan, A.M., Ismail, I., Gee, C.S., 2017. Mutual funds and stock market volatility: An empirical analysis of Asian emerging markets. *Emerg. Mark. Rev.* 31, 176–192.
- Rockafellar, R.T., Uryasev, S., 2000. Optimization of conditional value-at-risk. *J. Risk* 2, 21–41.
- Salisu, A.A., Olaniran, A., Tchankam, J.P., 2022. Oil tail risk and the tail risk of the US Dollar exchange rates. *Energy Econ.* 109, 105960.
- Scaillet, O., Treccani, A., Trevisan, C., 2020. High-frequency jump analysis of the bitcoin market. *J. Financ. Econom.* 18 (2), 209–232.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O., 2017. Proximal policy optimization algorithms. [arXiv:1707.06347](https://arxiv.org/abs/1707.06347).
- Shahzad, S.J.H., Bouri, E., Roubaud, D., Kristoufek, L., 2020. Safe haven, hedge and diversification for G7 stock markets: Gold versus bitcoin. *Econ. Model.* 87, 212–224.
- Singh, A., 2021. Investigating the dynamic relationship between litigation funding, gold, bitcoin and the stock market: The case of Australia. *Econ. Model.* 97, 45–57.
- Uryasev, S., 2000. Introduction to the theory of probabilistic functions and percentiles (value-at-risk). In: Uryasev, S. (Ed.), *Probabilistic Constrained Optimization. In: Nonconvex Optimization and Its Applications*, vol. 49, Springer US, pp. 1–25.
- Wei, W.C., 2018. The impact of Tether grants on Bitcoin. *Econom. Lett.* 171, 19–22.
- Zhang, C., Chen, H., Peng, Z., 2022a. Does bitcoin futures trading reduce the normal and jump volatility in the spot market? Evidence from GARCH-jump models. *Finance Res. Lett.* 102777.
- Zhang, Z., Shahzad, S.J.H., Bouri, E., 2022b. Tail risk transmission from commodity prices to sovereign risk of emerging economies. *Resour. Policy* 78, 102869.